

Network Defense and Secure Wi-Fi

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오네스

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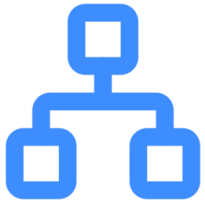
A guest joins the office Wi-Fi

Guests need the Internet.
Staff need private files.

Should both groups reach the same devices?

Today’s lesson

1. Firewall rules and segmentation	A connection test and an access rule
2. IDS/IPS, traffic evidence, VPNs	A packet finding and a protection boundary
3. Secure Wi-Fi	Practice choosing secure router settings



1. Control connections

Firewalls, filtering, and separate network groups

A firewall applies traffic rules

A firewall allows or blocks connections using rules.

Direction at your laptop	Meaning	Everyday example
Outbound ↗	Your laptop starts a connection.	Your browser opens the LMS.
Inbound ↙	Another device starts a connection to you.	A classmate connects to your shared folder.

A reply to your web request usually belongs to an existing allowed connection.

A useful rule names the traffic

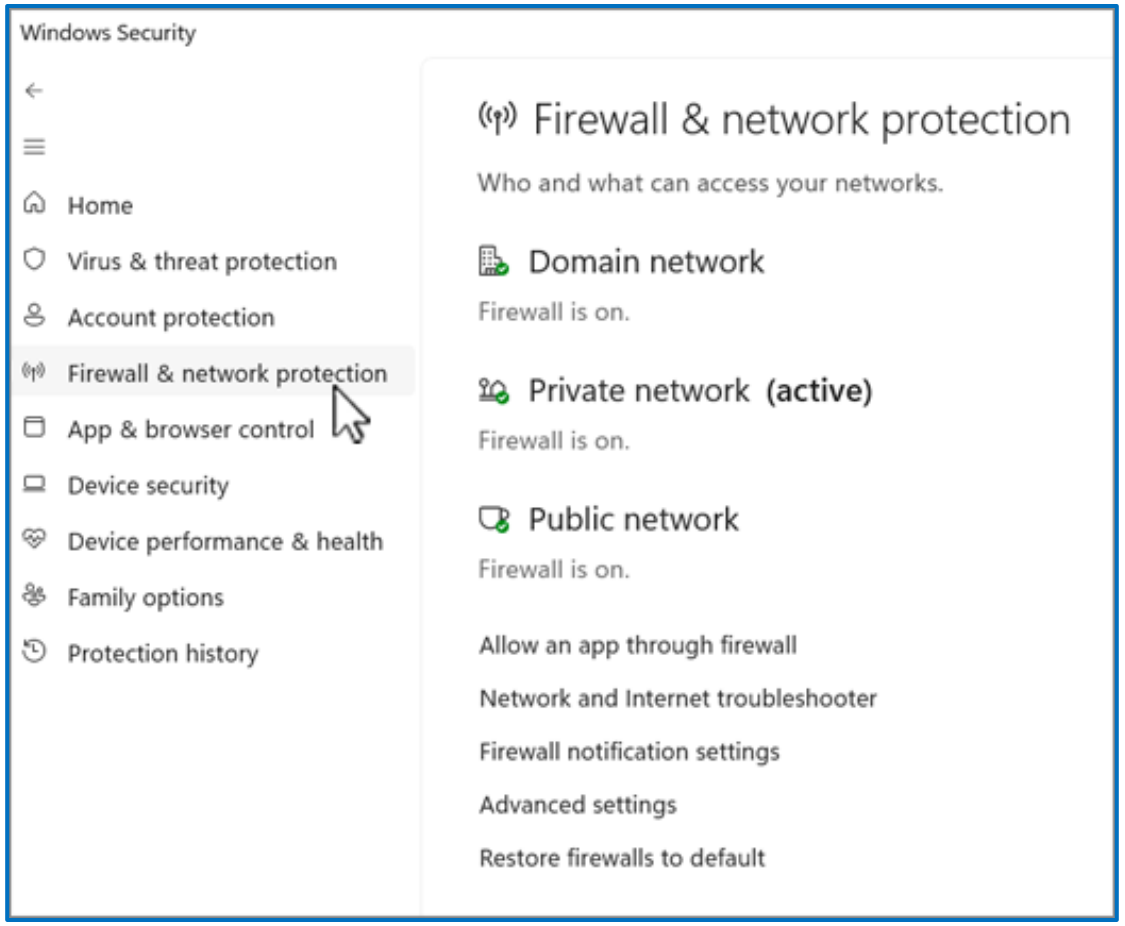
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Who starts it? Where does it go? Which service? Allow or deny?

Source	Destination and service	Decision
Guest network	Internet web browsing	Allow required access
Guest network	Staff payroll server, any service	Deny
Employee	Accessing payroll	Allow

Access needs an explicit reason. These are examples.

Inspect your firewall



1 Open Windows Security.

2 Choose Firewall & network protection.

3 Find the **active** network profile. Check that the firewall is on.

Activity 1: Record a working connection

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On an authorized Windows PC, open Command Prompt.

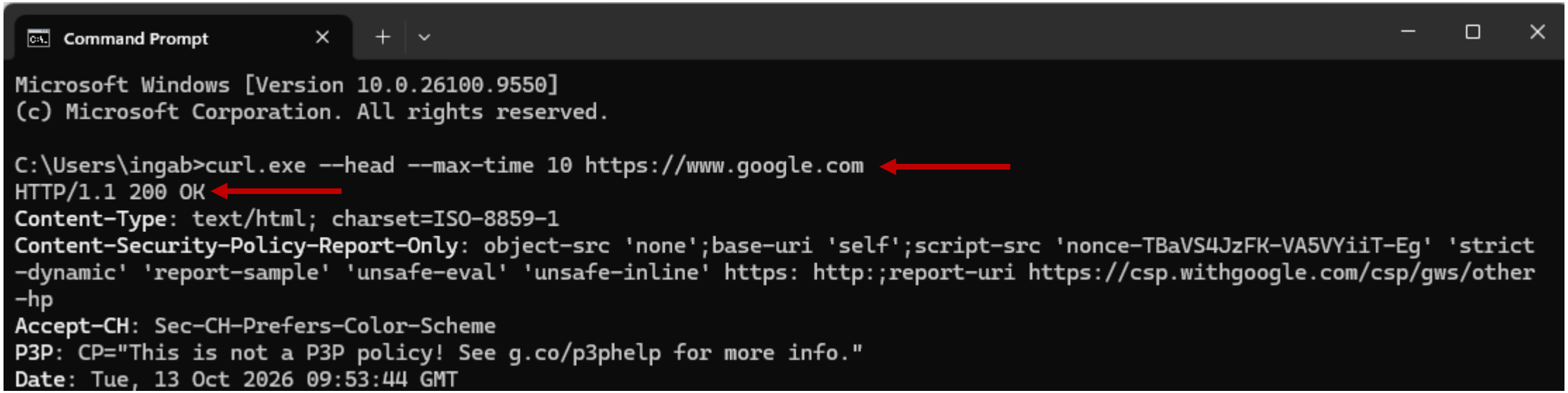
Type this as one line:

"%SystemRoot%\System32\curl.exe" --head --max-time 10 https://google.com

What it does	Expected observation
curl requests response headers.	HTTP response, e.g. HTTP/1.1 200 OK.
--max-time 10 caps the wait.	Stops within about 10 seconds.

Continue only if this baseline works. Save the result as “before”.

Activity 1: Record a working connection



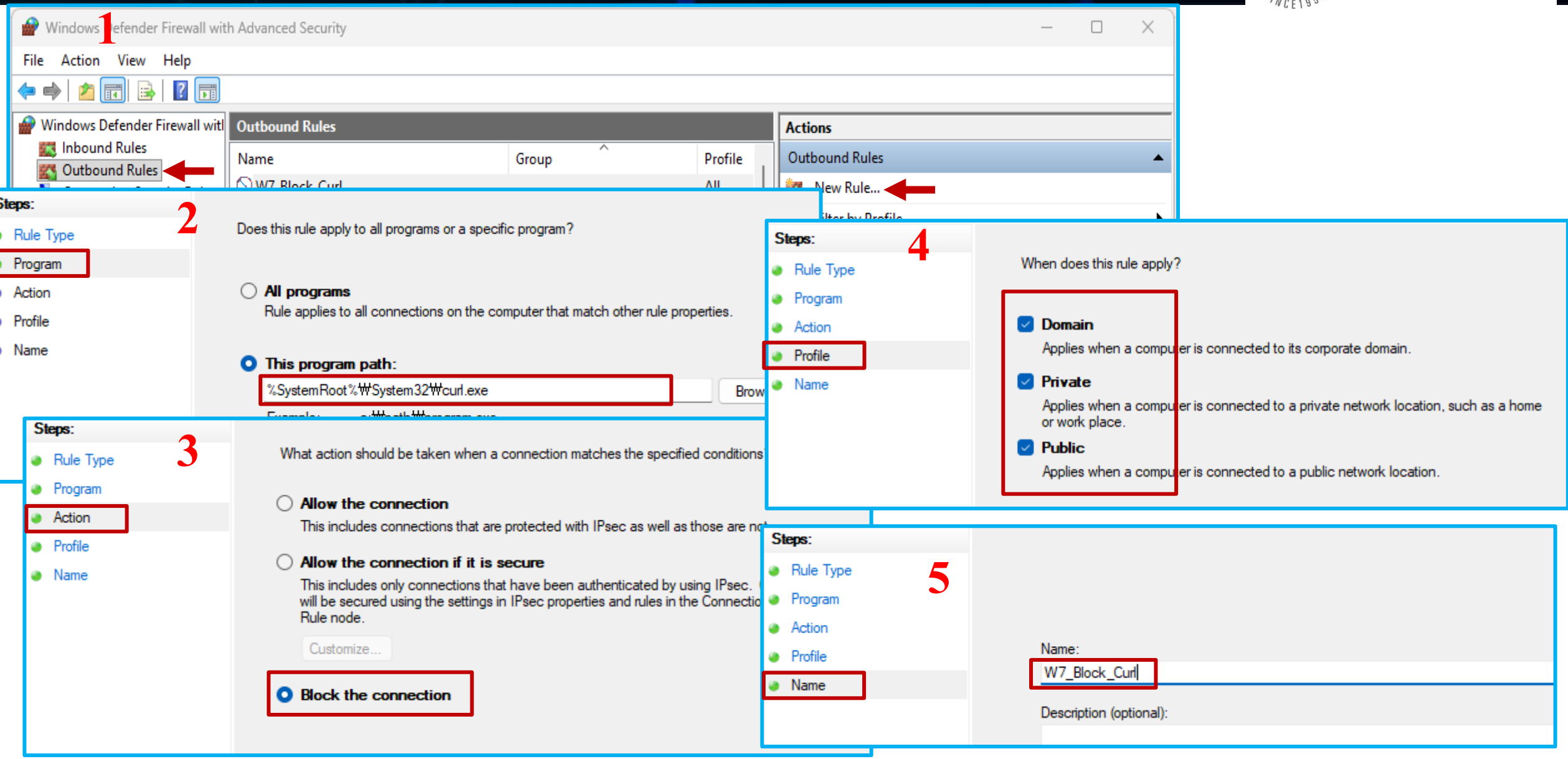
Activity 1: Block only the test program

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1. Press **Windows + R**, type **wf.msc**, and press **Enter**.
2. Select **Outbound Rules**, then **New Rule**. **Choose Program**.
3. **Program path**: %SystemRoot%\System32\curl.exe
4. Choose **Block the connection**. Select all profiles.
5. Name it **W7_Block_Curl**, then Finish.

Administrator rights required. Keep all existing rules unchanged.

Activity 1: Block only the test program: screenshots



Activity 1: Compare the result

Run the same command again.

"%SystemRoot%\System32\curl.exe" --head --max-time 10 https://google.com

Test	Expected result
curl after the new rule	curl cannot complete the connection after the rule is applied
The same site in your browser	The page can still open. The rule targets curl.exe.



Save the “blocked” result. Which program did the rule affect?

Activity 1: Undo and verify

1. In Outbound Rules, find W7_Block_Curl.
2. Delete only that rule.
3. Run the original command again.
4. Check that a web response returns. Save “after”.

```
C:\Users\ingab>curl.exe --head --max-time 10 https://www.google.com
curl: (7) Failed to connect to www.google.com:443 after 62 ms: Could not connect to server

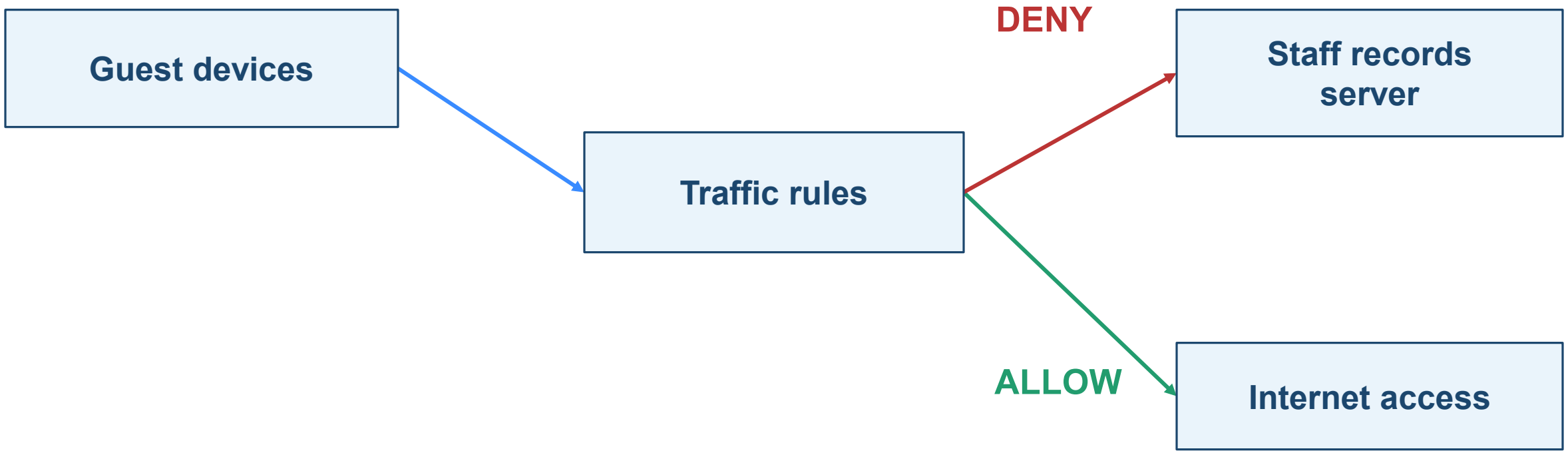
C:\Users\ingab>curl.exe --head --max-time 10 https://www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
```

A complete experiment shows: working, blocked, then working again.

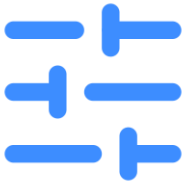
Segmentation limits access between groups

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Network segmentation means separating devices or services into groups and controlling the communications between them.



The guest can access the Internet but cannot access the staff records server.



2. Observe and protect traffic

IDS, IPS, Wireshark, and encrypted connections

IDS and IPS: detection and response

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Term	Meaning	Office example
IDS Intrusion Detection System	Watches for suspicious activity and raises an alert.	Repeated unusual connections trigger a warning.
IPS Intrusion Prevention System	Can act on detections to block or drop traffic.	An inline sensor drops traffic matching a known exploit.

An alert needs context

Example

Time	Evidence	Question
10:00	Staff laptop opens payroll using HTTPS.	Expected work?
10:01	Guest device tries several staff service ports.	Approved test or unwanted probing?
10:02	The sensor raises “unusual connection pattern”.	What should you verify first?

Choose one investigation step and one appropriate response.

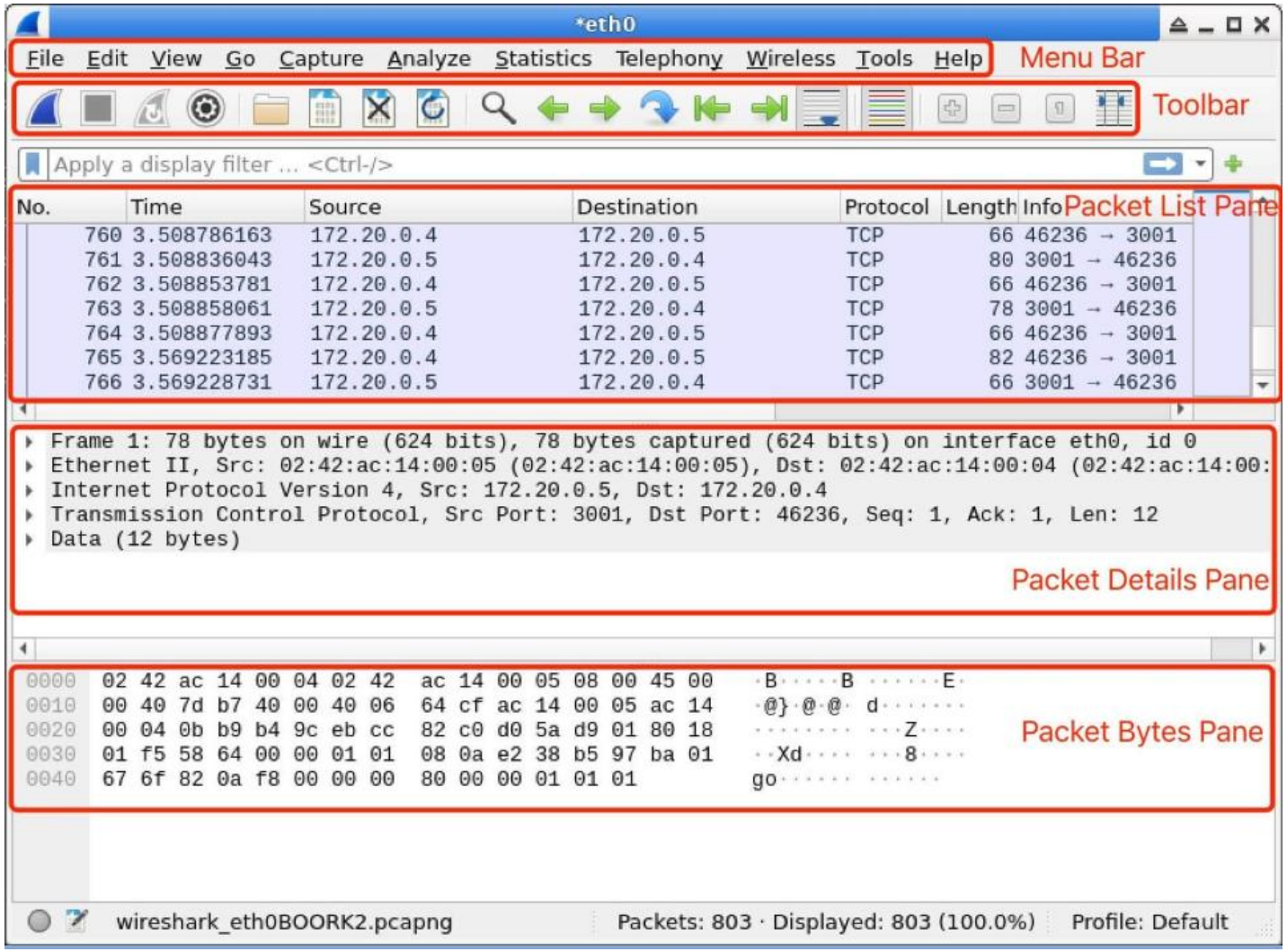
Detection can miss events or raise false alarms

Term	Plain meaning	Example
False positive	An alert for harmless activity.	An approved scanner triggers a warning.
False negative	Harmful activity produces no alert.	An unfamiliar attack avoids the detection rule.

A reasonable response to the previous log

Verify the source and permission. Check the guest boundary. Preserve evidence and report the finding.

Wireshark makes packet evidence visible



Top: packet list

Who sent data to whom?

Middle: packet details

Shows information contained in the selected packet, organized by protocol

Bottom: Packet bytes

Shows the raw captured bytes

Documentation example. The next lab uses a different capture.

Activity 2: Open a prepared capture

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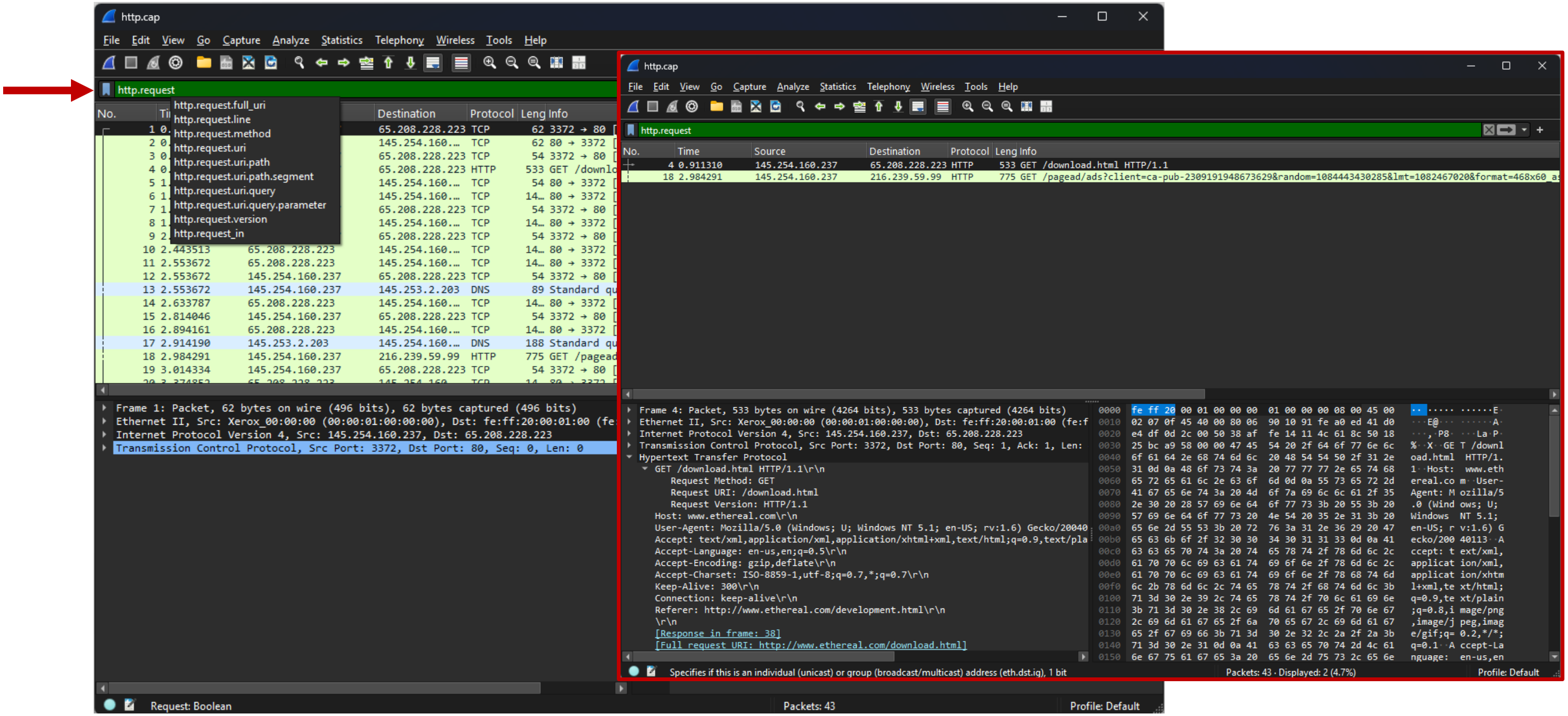
Install Wireshark from wireshark.org/download.html
https://www.wireshark.org/download.html

Download the official http.cap sample
https://wiki.wireshark.org/uploads/27707187aeb30df68e70c8fb9d614981/http.cap

1. Download the linked **http.cap** sample.
2. In Wireshark, **choose File** > **Open** and select **http.cap.**
3. Use the display filter bar at the top. Type **http.request** and press **Enter**.

A capture file contains recorded packets.

Activity 2: Open a prepared capture

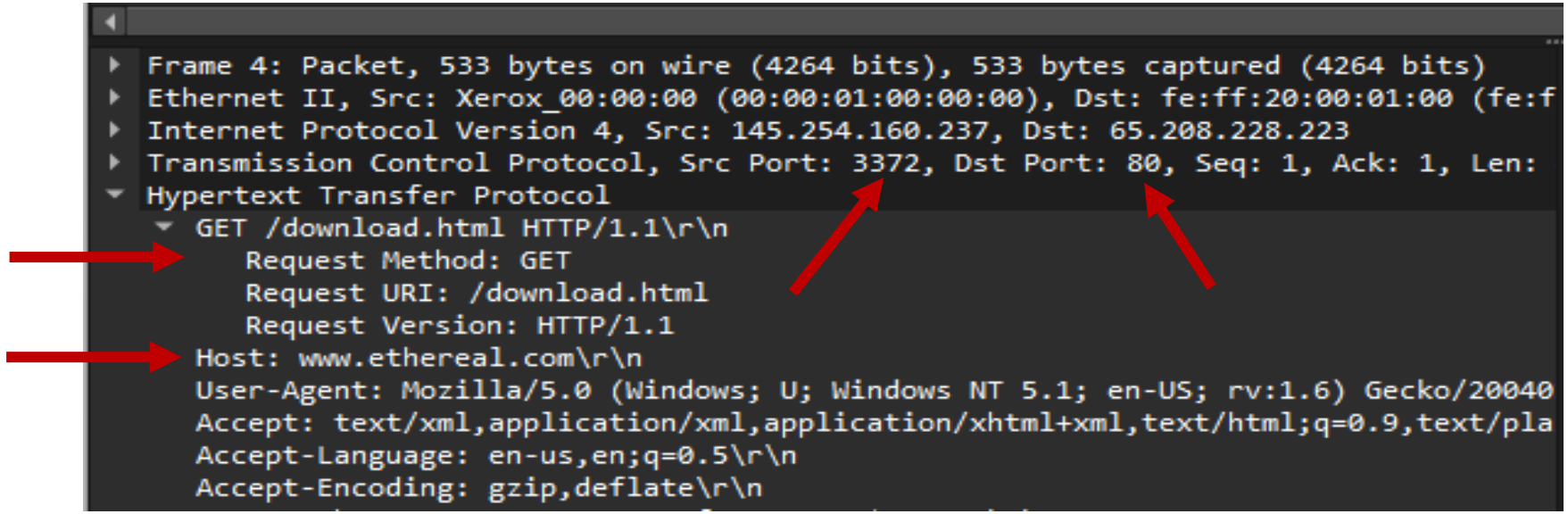


Activity 2: read the first HTTP request

Filter: http.request

1 Select the request for /download.html.

2 Expand Hypertext Transfer Protocol. Find the method and Host field.



Record

Method
Host name
Requested path
Destination port

Which part of this request could a network observer read?

Activity 2: expected evidence

Field	Value	Meaning
Method	GET	Request a resource
Host	www.ethereal.com	Named web server
Path	/download.html	Requested resource
Destination TCP port	80	Standard HTTP port

The capture exposes readable HTTP request data.

With correctly configured HTTPS, a network observer normally cannot read this application content.

Encryption has a protection boundary

Protection	Protected connection	What remains outside it
WPA2 / WPA3	Your device to the Wi-Fi access point.	Other parts of the Internet path.
HTTPS with TLS	Your browser to the TLS endpoint for the site.	Data on a compromised endpoint.
Encrypted VPN	Your device to the VPN gateway, for traffic using that tunnel.	Traffic beyond the gateway still needs protection.

These protections can work together.

A VPN protects a tunnel to its gateway

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Example: a remote worker connects to an approved office VPN(Virtual Private Network).



Encrypted tunnel: laptop to VPN gateway.

The office service still needs login permissions and HTTPS when applicable.

Which protection fits the problem?

A. A guest must not reach payroll.

B. An analyst needs to notice unusual traffic.

C. A student submits work over café Wi-Fi.

Situation	Suitable control	Important limit
Guest to payroll	Separate guest access and enforce a deny rule.	A different Wi-Fi name alone is insufficient.
Unusual connections	IDS alerting and investigation.	An alert can be wrong.
Course upload in a café	Valid HTTPS to the intended website.	HTTPS does not prove the site is honest.



3. Secure Wi-Fi

A router has two different passwords



Wi-Fi password

Allows a device to join the wireless network.

Router admin password

Allows an authorized person to change router settings.

Use different, strong passwords.

Choosing a Wi-Fi security mode

Mode	Plain meaning	Decision
WPA3-Personal	Modern protection for a personal network.	Prefer when all devices support it.
WPA2/WPA3 Transitional	New devices use WPA3. Older ones can use WPA2.	A compatibility option.
WPA2-Personal with AES	An older supported option using AES encryption.	Use when WPA3 is unavailable.
WEP, WPA/TKIP, or Open	Old weak protection, or no Wi-Fi encryption.	Avoid for a private home network.

Campus Enterprise Wi-Fi uses a different sign-in system. Follow the IT setup guide.

Router hardening reduces unnecessary access

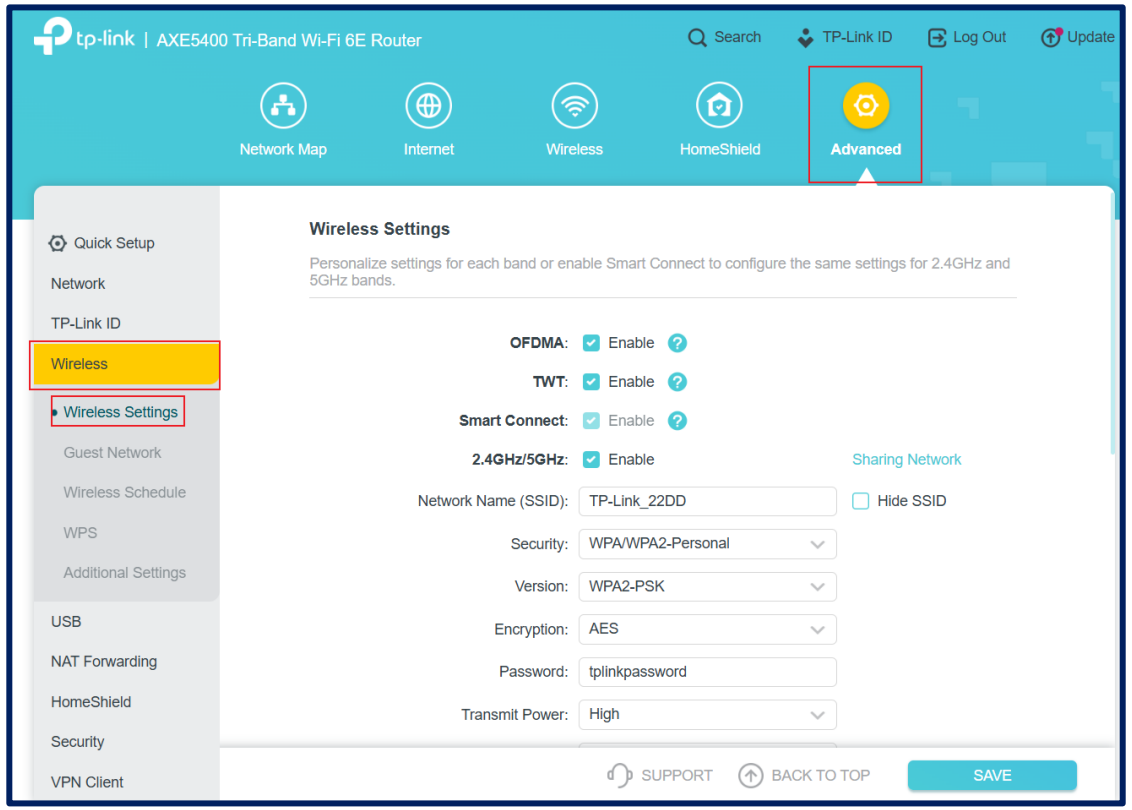
Setting	Practical choice	Why it helps
Admin access	Unique admin password. Disable remote admin if unused.	Limits who can change the router settings.
Firmware	Install supported vendor updates.	Fixes known software weaknesses.
WPS	Disable if you do not need it.	Removes an extra joining method.

Firmware = software inside the router. WPS = Wi-Fi Protected Setup.

Activity 3: a router in your browser

[Open the TP-Link Archer AX55 router emulator
https://emulator.tp-link.com/ax55v4_router/index.html](https://emulator.tp-link.com/ax55v4_router/index.html)





1 **Open Advanced.**

2 **Find Wireless.**

3 **Open Wireless Settings.**

Vendor example. Menus vary by model and firmware.

Activity 3: choose the wireless settings

In the emulator, practice on one enabled wireless band.

Field	Lab value or action
Network Name (SSID)	StudyLab_Staff
Security	WPA3-Personal if available. Otherwise WPA2-PSK with AES.
Password	Invent a long practice passphrase. Never use a real one.
Save	Save if supported. Record the selected values.

SSID means the advertised Wi-Fi name.

Thank You